

The empirical recurrence for OEIS A297523: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A297523 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 135$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A297523 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1 or 3 neighboring 1s.”. It has offset 1 and begins

1, 55, 727, 3383, 19187, 193087, 1671315, 11283382, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Dec 31 2017 09:57:52, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 135$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 135$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 270$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	1	c_{23}	−1157811923289	c_{45}	−433284746193606	c_{67}	161059079767380
c_2	16	c_{24}	−954662092330	c_{46}	2822358799449802	c_{68}	−47577866689576
c_3	109	c_{25}	700512655466	c_{47}	3748990242310340	c_{69}	−64937966648312
c_4	1498	c_{26}	8428815743100	c_{48}	3369345209564599	c_{70}	−7733066730032
c_5	1210	c_{27}	12256857466028	c_{49}	1689339732885711	c_{71}	−25218024415672
c_6	−11613	c_{28}	6758748705719	c_{50}	−4336481199156410	c_{72}	1275458302784
c_7	−56542	c_{29}	−11004842490048	c_{51}	−3346934497592033	c_{73}	2557173262400
c_8	−529021	c_{30}	−63540273275420	c_{52}	−4746627182231782	c_{74}	615727605280
c_9	−597479	c_{31}	−67666980665011	c_{53}	−3007296978322555	c_{75}	1815006248192
c_{10}	3559597	c_{32}	−30856765149359	c_{54}	3284178087986092	c_{76}	230401751296
c_{11}	11270952	c_{33}	62052380705002	c_{55}	1045651581223386	c_{77}	−22046171648
c_{12}	64903479	c_{34}	261613657585902	c_{56}	3655909192524071	c_{78}	68785995520
c_{13}	74117941	c_{35}	161625926351820	c_{57}	3004819863953915	c_{79}	−61671662080
c_{14}	−444983429	c_{36}	84262795161718	c_{58}	−1351729409469086	c_{80}	1029614080
c_{15}	−1117472726	c_{37}	−154554460533450	c_{59}	365089629738646	c_{81}	6257965056
c_{16}	−3377306952	c_{38}	−462721812803053	c_{60}	−1604813751493994	c_{82}	−918441984
c_{17}	−3150931089	c_{39}	147548826608038	c_{61}	−1748743222933420	c_{83}	294658048
c_{18}	24543625808	c_{40}	−2856872961478	c_{62}	265659562626554	c_{84}	75382784
c_{19}	54303953049	c_{41}	176209747607767	c_{63}	−434226915772566	c_{85}	−30212096
c_{20}	77828446786	c_{42}	−371601738239197	c_{64}	380267335014984	c_{86}	−65536
c_{21}	17716745644	c_{43}	−1726145629984786	c_{65}	522936775633800	c_{87}	262144
c_{22}	−622893082067	c_{44}	−1007592982631577	c_{66}	−21632438636004		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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